Operating Conditions

Rotor operating stages	
forward velocity, V	148 knots
tip speed, ΩR	229 m/sec (750 fps)
shaft angle-of-attack, α_s	0
advance ratio, μ	0.333
advancing tip Mach number, $M_{1.0,90}$	0.895
Rotor force trim (target values)	
thrust, T	73,395 N (16,500 lb)
propulsive force, PF	8260 N (1857 lb)
side force, Y	0
helicopter parasite drag area, f	2.32 m^2 (25 ft^2)
rotor blade loading, C_T/σ	0.090

Calculation Parameters

Blade motion calculated by a harmonic analysis method.

azimuth increment	5°
number of harmonics	15
tolerance on blade mode amplitude and velocity, in test for convergence	0.001
number of blade segments (radial)	15

Iterate on $\theta_{.75}$, A_{1S}, and B_{1S} to trim rotor thrust, propulsive force, and side fo tolerances:

thrust	445 N
propulsive force	44 N
side force	44 N